

Discussion of: Fan Charts 2.0: Flexible Forecast Distributions with Expert Judgement

A. Sokol

Andrea De Polis¹

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¹University of Strathclyde

The paper in one slide

The paper proposes a new, univariate approach to perform *distributional* scenario analysis.

- Introduces Conditional Quantile Regression (CQR).
- Flexible tool to incorporate judgement in density forecasts.
- Out-of-sample performance at par with standard off-the-shelf models.
- Application on house price inflation response during the pandemic.

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The paper advances an interesting methodology with potential widespread applications, both for policy making and risk management more in general.

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CQR extends McCracken & McGillicuddy (2019) conditional forecasting framework by introducing the quantile dimension to the problem.

- CQR for direct forecasting (h)

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↪ Does this hold across quantiles, e.g., within a nonlinear setting?

Persistence

CQR considers changes in the conditional distributions, but it does not account for persistence in the higher-order moments

- Stochastic volatility is a well documented feature of the series modelled.
- Skewness as well, as I will show tomorrow (!!!)

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Assume that (similarly to Engle & Manganelli 2004)

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$$\vdots$$

$$q_t^{(\tau)} = \phi^{(\tau)t} q_0^{(\tau)} + \beta^{(\tau)} \sum_{j=0}^{t-1} \phi^{(\tau)j} X_{t-j}$$

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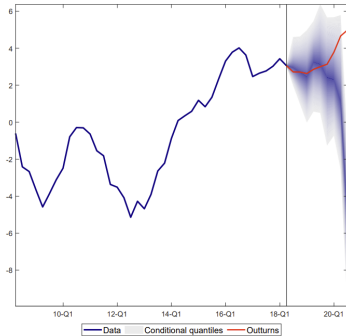
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If the *distribution* of y_t features autocorrelation, then a small number of lags of X might induce **truncation bias** in the dynamics of the fitted distributions.

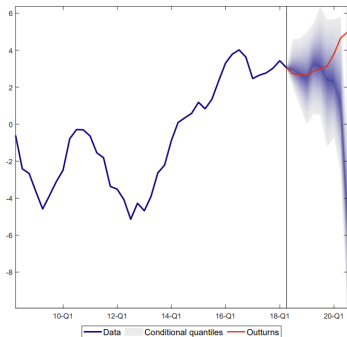
Conditional densities

CQR



Conditional densities - what's going on?

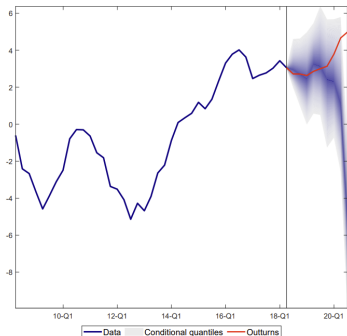
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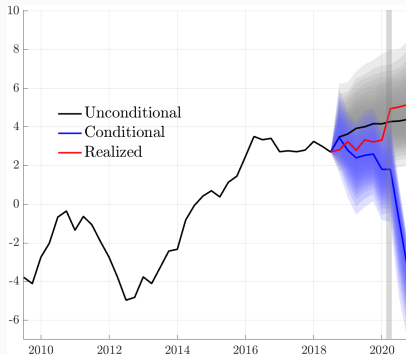
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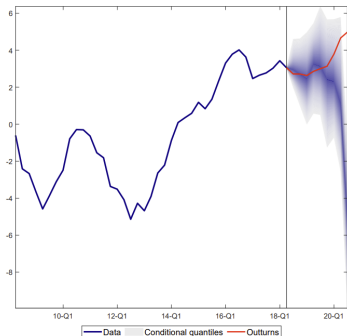
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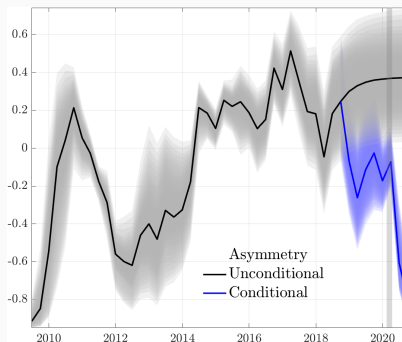
- What happens to the predictive moments?
- Delle Monache et al. (2024) can produce qualitatively similar results, relating technical assumptions to conditional moments.

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- What happens to the predictive moments?
- Delle Monache et al. (2024) can produce qualitatively similar results, relating technical assumptions to conditional moments.
- In this setting, **asymmetry** drops significantly, the model less affected, but the conditional scenario remain similar.

Out-of-sample and model comparison

When compared to an off-the-shelf BVAR for the same set of variables:

- What about point forecast?
- The CQR starts becoming competitive after 3/4 quarters in terms of density forecasts.

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- What about point forecast?
 - The CQR starts becoming competitive after 3/4 quarters in terms of density forecasts.
 - Is this comparison fair to the CQR model?
- ↪ Marcellino et al. (2006) and McCracken & McGillicuddy (2019) document that *iterated forecasts* retain some superior forecast ability compared to *direct forecasting* – for point predictions.

Wrapping up

- I think CQR is a very nice idea!
- It is a promising tool to complement risk managers/policy makers arsenal.
- I enjoyed the paper and I would like to see CQR applied to other case-studies to see the full potential of this approach.